Unit: 6 Unit 2: Exponents Radicals
Topic: Radicals

Name: ____________
Date: ________

Advanced (Carolina Reaper)

Q1. What is the simplified form of $\sqrt{16}$?
(A) 2
(B) 4
(C) 8
(D) 16
(E) 32

Q2. What is $\sqrt{9}$ in simplest radical form?
(A) 1
(B) 3
(C) $\sqrt{3}$
(D) $\sqrt{9}$
(E) 9

Q3. Simplify the expression $\sqrt{64}$.
(A) $\sqrt{16}$
(B) 8
(C) $\sqrt{32}$
(D) 16
(E) 32

Q4. What is $\sqrt[3]{27}$?
(A) 3
(B) 9
(C) 27
(D) $\sqrt{27}$
(E) $\sqrt[3]{9}$

Q5. What is the value of x in $x^2 = 16$ if x is positive?
(A) 2
(B) 4
(C) 8
(D) 16
(E) 32

Q6. Express $\sqrt{25}$ as a rational exponent.
(A) $5^{1/2}$
(B) $25^{1/2}$
(C) 5^2
(D) 25^2
(E) 5

Q7. Simplify $\sqrt{8}$.
(A) 2
(B) $2\sqrt{2}$
(C) 4
(D) $\sqrt{16}$
(E) $\sqrt{4}$

Q8. Convert $x^{2/3}$ into radical form.
(A) $\sqrt[2]{x}$
(B) $\sqrt[3]{x}$
(C) $\sqrt[3]{x^2}$
(D) $\sqrt{x}$
(E) $\sqrt{x^2}$

Open Ended Questions (FRQ)

Q9. Prove that $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$ and provide an example.

Q10. Explain how to convert between rational exponents and radical form, and provide two examples.

Chef's Tips

- Emphasize the rules of exponents and radicals to students before the test.
- Allow the use of calculators only for checking answers.
- Encourage students to show their work and explain their reasoning for Free Response Questions.